



GE Medical Systems

Technical Publications

Direction 5131778
Revision 4

ACGD to HFD-S Upgrade Installation

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Operating Documentation

DAMAGE IN TRANSPORTATION

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- Call 1-800-548-3366 and use option 8.
- Fill out a report on <http://3.28.216.127/sctq/InstallFulfill/IFHome.htm>
- Contact your local service coordinator for more information on this process.

Rev. 08/15/2003

Direction 2128126 - Language Policy For Service Documentation, Rev 1

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES EQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ETE CONSULTE ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAINER CHEZ LE TECHNICIEN, L'OPERATEUR OU LE PATIENT DES BLESSURES DUES A DES DANGERS ELECTRIQUES, MECANQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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注意：

- 本维修手册仅存有英文本。
- 非 GEMS 公司的维修员要求非英文本的维修手册时，客户需自行负责翻译。
- 未详细阅读和完全了解本手册之前，不得进行维修。
- 忽略本注意事项会对维修员，操作员或病人造成触电，机械伤害或其他伤害。

LIST OF EFFECTIVE PAGES

REV	DATE	PRIMARY REASON FOR CHANGE
1	Jan 11, 2005	Initial release.
2	Mar 18, 2005	Corrections After 12x M3A Software Release
3	Aug 17, 2005	Updated cable map on page 17 with Run numbers and part numbers for new cables.
4	Jun 5, 2008	Added gradient amplifier power supply replacement procedures (PSR 13111994).

PAGE	REV	PAGE	REV	PAGE	REV	PAGE	REV
Title Page.....	1	Title Page.....	2	Title Page.....	3	Title Page.....	4
Damage in Trans.	-	Damage in Trans	-	Damage in Trans	-	Damage in Trans	-
Direction 2128126 .1*		Direction 2128126		Direction 2128126		Direction 2128126	
5 to 15	1	1*		1*		1*	
		5 to 31.....	2	5 to 31.....	3	1 to 45.....	4

* This revision/letter corresponds to the indicated document's revision control system.

TABLE OF CONTENTS

1- OVERVIEW	7
2- REQUIREMENTS	7
2-1 Personnel.....	7
2-2 Tools and Test Equipment.....	7
2-3 Consumables.....	8
2-4 Replacement Parts	8
2-5 Required Conditions	8
2-7 Safety.....	8
3- HARDWARE INSTALLATION PROCEDURE	8
3-1 Preliminary Procedure Requirements	8
3-2 Power Down and LOTO	8
3-3 Modify the Fan in the Gradient Cabinet.....	8
3-4 SGA Control Board Replacement.....	9
3-3 Gradient Processor Replacement	10
3-4 Back Panel Replacement	11
3-5 CAN Communication Network Connection	15
3-6 MDS Link Reroute	16
3-7 Affix Labels	18
3-8 1.5T TRM Systems ONLY – Replace Gradient Filter.....	18
3-9 Gradient Amplifier Power Supply Replacement	19
3-10 LOTO Removal and Power-Up.....	32
4- SOFTWARE INSTALLATION AND MODIFICATIONS	33
5- FUNCTIONAL CHECKS.....	34
6- CALIBRATIONS	36
7- FINALIZATION	36
APPENDIX A – INVENTORY LIST.....	37
APPENDIX B – FAN MODIFICATION INSTRUCTIONS	38
B-1 EFFECTIVITY.....	38
B-2 FIELD SUPPLIED MATERIALS	39
B-3 PURPOSE.....	40
B-4 PROCEDURE	40
B-2.1 Ensure Proper LOTO.....	40
B-2.2 Replacement Procedure Steps.....	41
APPENDIX C – GOLDSEAL RETURN SHEET	45

1- OVERVIEW

The HFD-S is the upgrade gradient configuration for Eclipse upgrade to EXCITE (E2) systems at M4 and EXCITE HD systems at M3A software (and beyond).

The general flow of this installation procedure is as follows:

- Hardware Installation
 - Lock Out Tag Out
 - Modify Fan in the Gradient Cabinet (FMI 60682 if not already done)
 - Replace SGA Control Boards with HFA Control Boards
 - Replace the GP2 with the GP3
 - Replace the PDU Back Panel
 - Configure CAN Communication with the HFD-S
 - Reroute the MDS Link
 - Affix Label
 - (1.5T TRM Systems ONLY): Replace the Gradient Filter
 - Remove Lock Out Tag Out
- Software Installation
- Calibrations
- Functional Checks

2- REQUIREMENTS

2-1 Personnel

System Type	# FEs	Hours
TRM (1.5T and 3.0T)	1	13
BRM/CRM (1.5T and E2)	1	9

2-2 Tools and Test Equipment

Item	Qty	Description	Part Number
HFD-S Upgrade Kit	1	HFD-S Upgrade Kit	M3000PT (BRM/CRM/3.0T) -OR- M3000PW (1.5T TRM)
Phillips Screwdriver	1	#2 Phillips	
Flat Head Screw Drivers	2	#3 Flat head, #5 Flat head	
Nut Driver	1	Nut Driver 5mm nut driver	
Needle Nose Pliers	1	Needle nose pliers	
Dikes	1	Needed to cut screws	
Wrench	1	3/8 Inch	
3.0T Grafidy 3 Kit	1	ECMT-3T Fixture Assembly	2386042
1.5T Grafidy 3 Kit	1	Service Tool Kit – Grafidy II	46-307164G3

2-3 Consumables

None

2-4 Replacement Parts

No replacement parts required.

2-5 Required Conditions

Prior to beginning the upgrade to HFD-S, it is recommended that:

1. All system information (images, protocols, etc) and calibrations be saved.
2. The Field Engineer checks that site has received all items in inventory list, located in Appendix A.
3. For 1.5T TRM systems: Field Engineer has downloaded Gradient Filter replacement procedure from the Service Methods website or CD under "Replacements"-> "Penetration Panel" -> "Gradient Filter Replacement Procedure"

2-7 Safety



⚠ DANGER

ELECTROCUTION HAZARD!

LETHAL VOLTAGES ARE PRESENT WITHIN THE PDU.

CHECK THAT POWER AT THE MAIN DISCONNECT IS OFF, LOCKED, AND TAGGED BEFORE PROCEEDING.

3- HARDWARE INSTALLATION PROCEDURE

3-1 Preliminary Procedure Requirements

1. Confirm with the customer that all images that are on the system have been archived.
2. Perform a SaveINFO prior to beginning.

3-2 Power Down and LOTO

1. Notify all personnel in the area that you will be powering down the system to modify the ACGD.
2. Power down the computer and the whole system following the normal operation procedure for lock out tag out of the Main Disconnect Panel (MDP). If the site does not have an MDP, lock out tag out must be performed at the power source of the ACGD. *(For the remainder of this document, the main power source to the ACGD will be referred to as the MDP)*
3. Lock-Out Tag-Out the power at the MDP and verify power is off.

3-3 Modify the Fan in the Gradient Cabinet

1. Locate FMI 60682 instructions in Appendix B.

2. If this FMI has not already been performed at the site, follow the instructions in the FMI 60682 document to change the fan in the gradient cabinet.

3-4 SGA Control Board Replacement

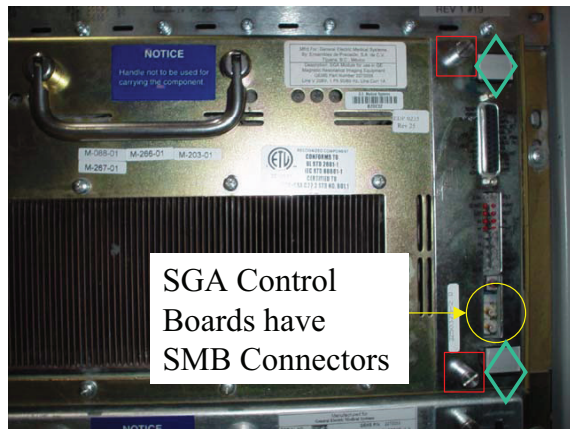


NOTICE

PREVENT PERMANENT DAMAGE TO THE STATIC-SENSITIVE BOARDS. ATTACH ANTI-STATIC WRIST STRAP TO YOUR WRIST AND TO A BARE METAL GROUNDING POINT,

1. Disconnect the 3 SGA control boards by loosening the screws, pushing the ejector tabs, then pulling the boards out. See Illustration 3-1. (Note the SGA boards are a **GREEN** color with SMB connectors on the front.)
2. Insert the 3 new HFA control boards until they lock in place. Tighten the screws on all three. See Illustration 3-1. (Note the HFA control boards are **BLUE** in color and have BNC connectors on the front) **DO NOT MIX THE BOARDS WITHIN THE CABINET**

NOTE: The HFD-S requires the use of HFA (**BLUE**) control circuit boards (with BNC connectors on the front). Using SGA (**GREEN**) control circuit boards (with SMB connectors on the front) will result in less than optimal gradient performance!!



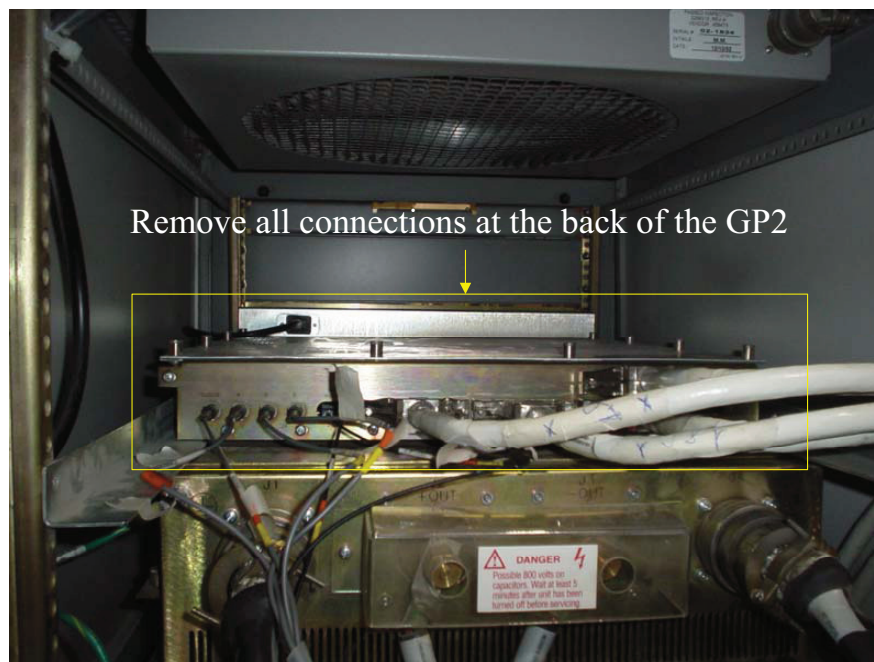
1. Loosen the Screws (□)
2. Push the Ejector Tabs (◇)
3. Pull out the 3 SGA Boards
4. Replace 3 SGA boards with 3 HFA Boards
5. Tighten the Screws on the HFA boards



REMOVAL OF SGA BOARDS AND INSTALLATION OF HFA BOARDS
ILLUSTRATION 3-1

3-3 Gradient Processor Replacement

1. At the back of the GP2 (inside the back cover of the gradient cabinet) disconnect the following, see Illustration 3-2:
 - a) MDS link
 - b) Four (4) fiber optic cables for the waveforms (clock, x, y, and z)
 - c) Four (4) 37-pin gradient control cable connections
 - d) Power cord



REMOVE ALL CONNECTIONS AT THE BACK OF THE GP2
ILLUSTRATION 3-2

2. At the front of the gradient cabinet, disconnect the following:
 - a) GSC cable (disconnect cable at both ends and discard)
 - b) Four mounting screws on the GP2.
3. Pull the GP2 out.
4. Insert the new GP3 half way, install the power connection and push it in the rest of the way. Connect the four mounting screws.
5. At the back of the gradient cabinet reconnect the following to the GP3:
 - a) Four (4) fiber optic cables for the waveforms (clock, x, y, and z)
 - b) Four (4) 37-pin gradient control cable connections
 - c) Power cord
 - d) **DO NOT** reconnect the MDS link!!!! It will no longer be used to communicate with the HFD-S.

- e) Connect the two (2) CAN cables from the HFD-S upgrade kit to the back of the GP3 at connections J7 and J8. Snake the 2 CAN cables down the left side of the cabinet (outside of the rail blocks of the amplifiers, and away from the black power leads) and feed them through the PDU hole in the left side of the cabinet. These two cables will be connected in the following section "Back Panel Replacement".
- f) (TRM Systems ONLY) Connect the one (1) Gradient Switch cable from the HFD-S upgrade kit to the back of the GP3 at connection J9. Snake this cable down the left side of the cabinet along with the two CAN cables. This cable will be connected in the following section "Back Panel Replacement"

3-4 Back Panel Replacement

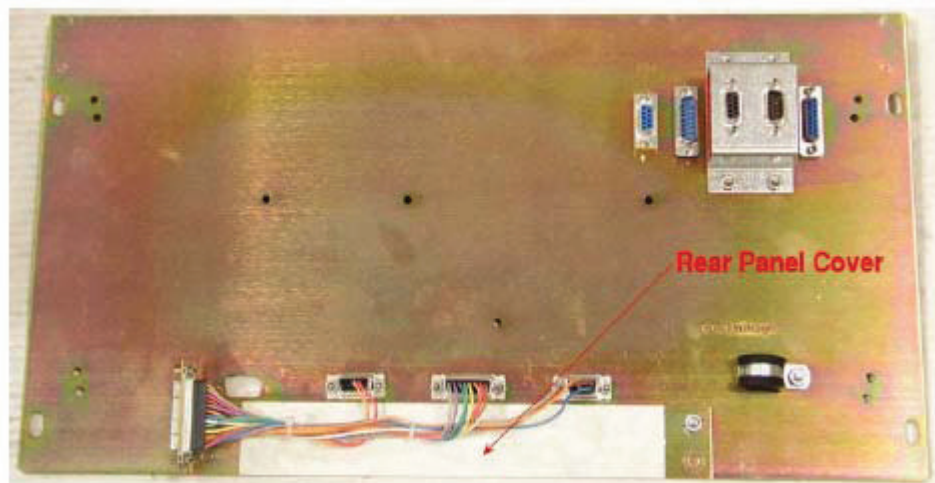
⚠ DANGER



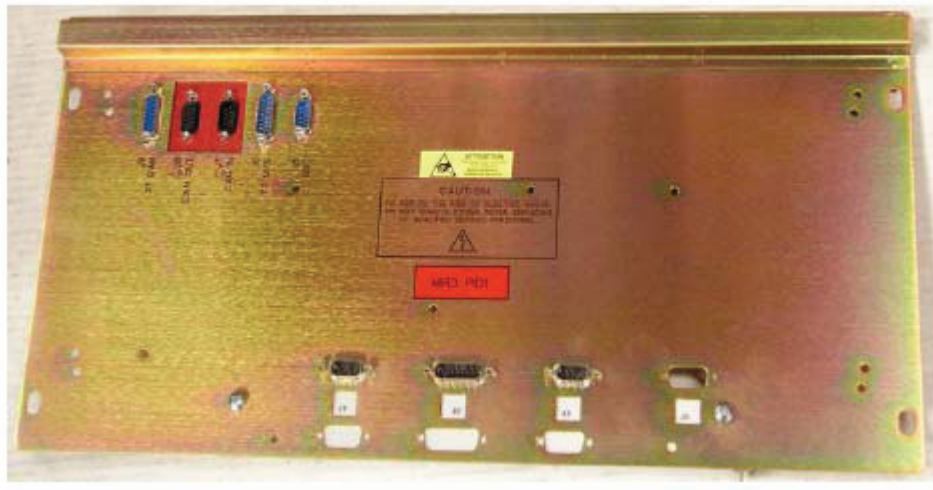
ELECTROCUTION HAZARD!

LETHAL VOLTAGES ARE PRESENT ON THE CIRCUIT BOARD ON THE INSIDE OF THE ACGD'S UPPER BACK PANEL. BE SURE SITE POWER HAS BEEN LOCKED OUT AND TAGGED OUT PRIOR TO BEGINNING THIS PORTION OF THE UPGRADE

1. Review Illustrations 3-3, 3-4, and 3-5 to determine which style PDU back panel is on site, ACGD Style 1, ACGD Style 2, or G3 Style. Follow the instructions relevant to the PDU at this site.
2. ACGD STYLE 1 (See Illustration 3-3):
 - a) Make sure proper Lock out / Tag out at the source of power to the PDU
 - b) Remove the rear panel mounting screws. Set aside for use during re-assembly.
 - c) Disconnect cables to the rear panel. Note connection location(s) for re-assembly.
 - d) Remove connector assembly from rear panel
 - e) Place rear panel cover in the proper position
 - f) Re-assemble connector assembly on replacement rear panel.
 - g) Connect the two (2) CAN cables from the GP3 (previously fed through the hole in the left side of the ACGD cabinet PDU Cover) to their respective 9-pin pass through connection points at J7 and J8 the inside of the new sheet metal back panel. Make sure you connect them to the **inside** connections of the back panel.
 - h) (TRM Systems ONLY) Connect the one (1) Gradient Switch Cable to J5 at the inside of the new sheet metal back panel. Make sure you connect it to the **inside** connection of the back panel.
 - i) Mount rear panel assembly on unit.



Note position of rear panel cover

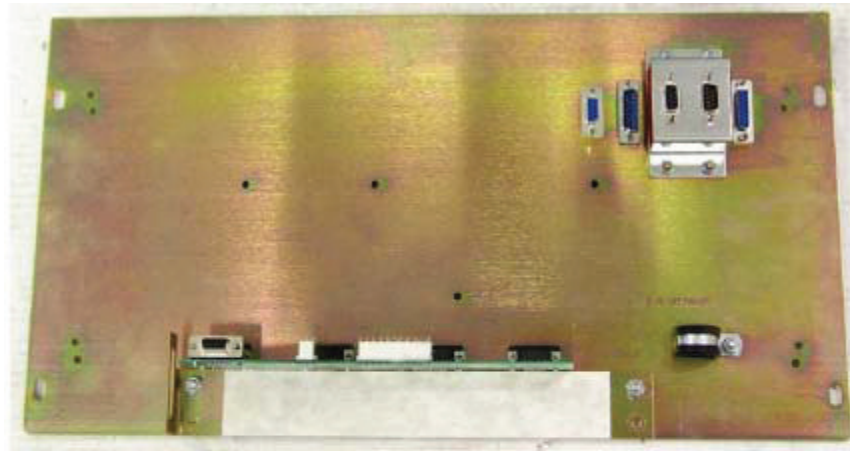


ACGD STYLE 1
ILLUSTRATION 3-3

3. ACGD STYLE 2 (See Illustration 3-4):

- Make sure proper Lock out / Tag out at the source of power to the PDU
- Remove rear panel mounting screws and set aside for use during re-assembly.
- Disconnect cables to the rear panel. Note connection location(s) for re-assembly.
- Remove connector assembly from rear panel.
- Place rear panel cover in proper position.
- Re-assemble connector assembly on replacement rear panel.
- Connect the two (2) CAN cables from the GP3 (previously fed through the hole in the left side of the ACGD cabinet PDU Cover) to their respective 9-pin pass through connection points at J7 and J8 the inside of the new sheet metal back panel. Make sure you connect them to the **inside** connections of the back panel.

- h) (TRM Systems ONLY) Connect the one (1) Gradient Switch Cable to J5 at the inside of the new sheet metal back panel. Make sure you connect it to the **inside** connection of the back panel.
- i) Mount rear panel assembly on the unit.



Note position of rear panel cover



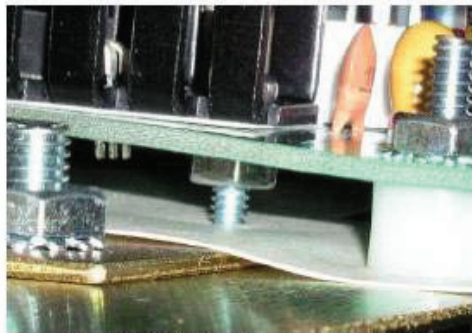
ACGD STYLE 2 ILLUSTRATION 3-4

4. G3 STYLE (See Illustration 3-5):

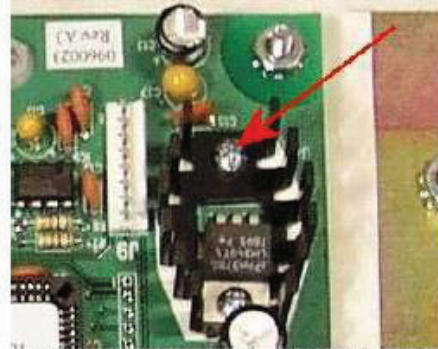
- a) Ensure proper Lock out / Tag out at the source of power to the PDU
- b) Remove rear panel mounting screws. Set aside for use during reassembly.
- c) Disconnect cables to rear panel. Note connection location(s) for reassembly.
- d) Remove PC Board assembly from rear panel.
- e) Place rear panel cover in proper position.
- f) Re-assemble PC Board assembly on replacement rear panel.

Note: It may be necessary to shorten the 4-40 x 3/8 inch screw on the PC Board that may hit the rear panel cover. Refer to photos showing the screw that has been cut shorter. If replacing the screw, use 4-40 x 1/4 inch and torque to 3 in/lbs. Only do this step if the screw is too long and hits the rear panel cover.

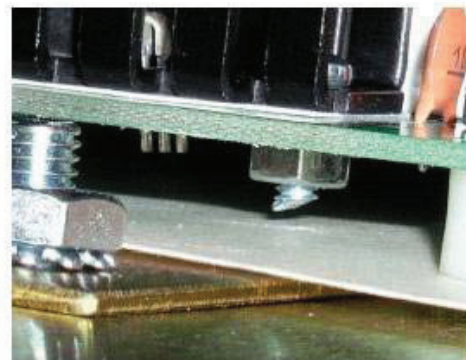
- g) Connect the two (2) CAN cables from the GP3 (previously fed through the hole in the left side of the ACGD cabinet PDU Cover) to their respective 9-pin pass through connection points at J7 and J8 the inside of the new sheet metal back panel. Make sure you connect them to the **inside** connections of the back panel.
- h) (TRM Systems ONLY) Connect the one (1) Gradient Switch Cable to J5 at the inside of the new sheet metal back panel. Make sure you connect it to the **inside** connection of the back panel.
- i) Mount rear panel assembly on unit.



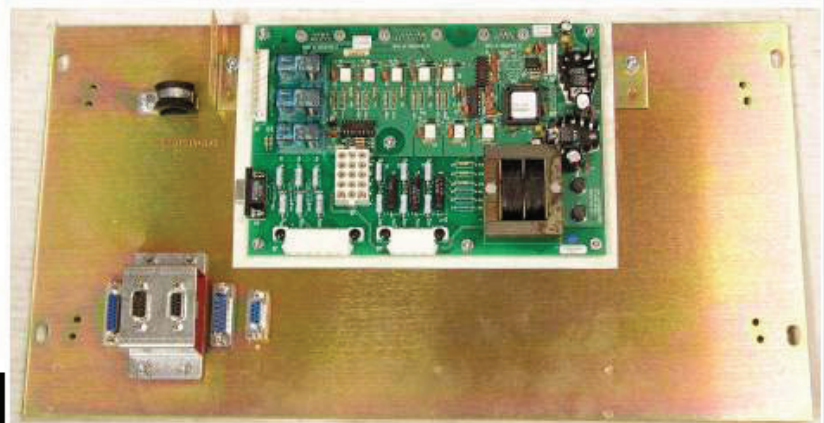
Screw shown hitting rear panel cover



Location of screw that may require shortening



Screw after shortening procedure



Note position of rear panel cover.

G3 STYLE BACK PANEL REMOVE CIRCUIT BOARD FROM OLD BACK PANEL AND INSTALL ON NEW BACK PANEL

ILLUSTRATION 3-5

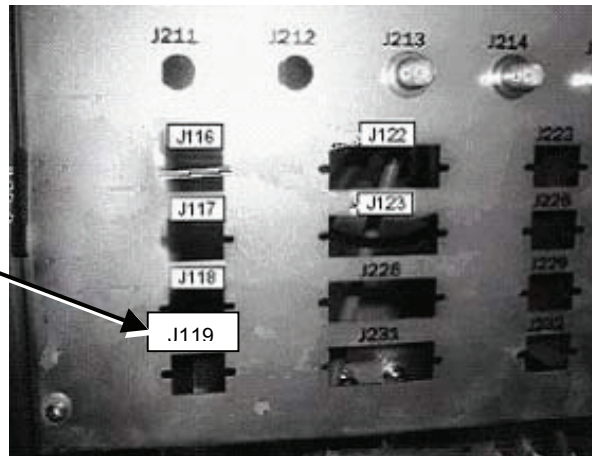
3-5 CAN Communication Network Connection

There are different CAN connections for 1.5T and 3.0T systems. Follow the direction specific to your system.

1. 3.0T Systems

- a) For sites that have been upgraded to EXCITE from Eclipse (E2), locate the label J119. Attach label to an empty slot in the I/O panel at the back of the Accessories Cabinet. See Illustration 3-6.

Attach label J119
to back panel of
Accessories Cabinet
(E2 systems only)



E2 ONLY: ATTACH LABEL J119 TO THE E2 ACC CABINET
ILLUSTRATION 3-6

- b) E2 ONLY: Route the CAN cable down the side of the accessories cabinet to the I/O Panel slot J119. Insert and attach the sub-D connector to the slot labeled J119.
- c) Connect Run 1263 from J119 at the I/O panel of the accessories cabinet to the back panel of the ACGD J7.
- d) Follow the table in Illustration 3-7 for information on how to connect the CAN communication links.

2. 1.5T Systems

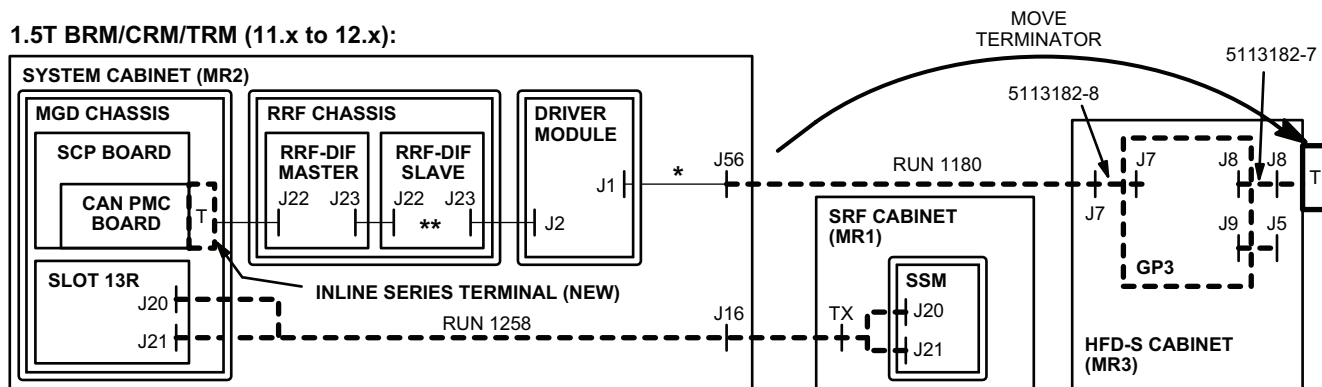
- a) Follow the table in Illustration 3-7 for information on how to connect the CAN communication links.

3-6 MDS Link Reroute

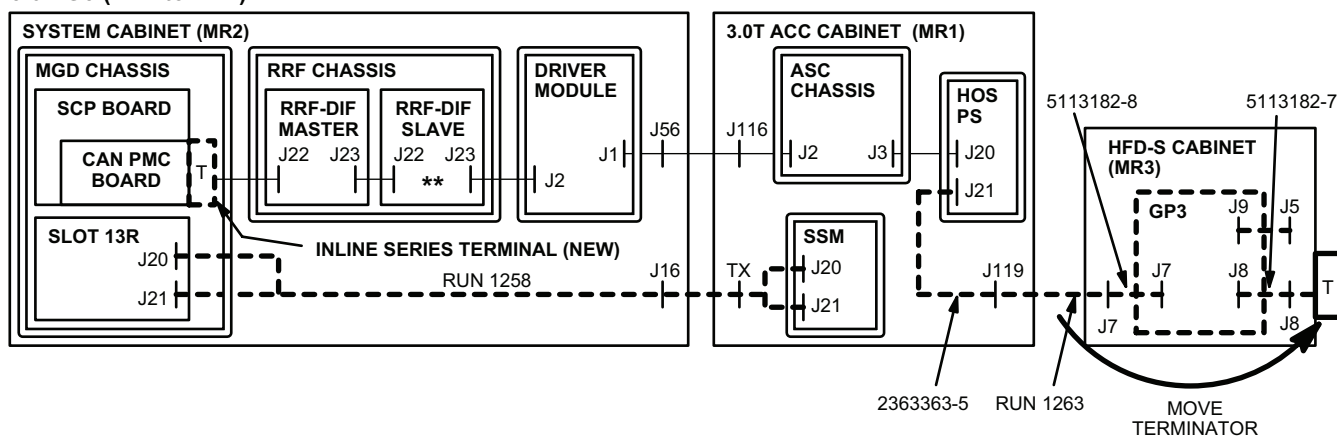
This section gives instructions on how to re-route the fibers of the MDS link from the SSM to the STIF board as they will no longer be needed to communicate with the Gradient Processor (GP3) of the HFD-S. See Illustration 3-7.

1. On the SSM, find the fiber optic connection labeled "MDS". Disconnect the cables from this connection.
2. On the back of the STIF board, disconnect the fiber optic cables connected to J20 and J21.
3. The fiber optic cables that formerly ran between the STIF & GP and the GP & SSM will no longer be used. These are in a solid conduit and cannot be easily removed. Be sure to mark the ends of these cables as "UNUSED" to avoid confusion with the unused cables at a later date.
4. Reconfigure the MDS link per Illustration 3-7. Use the new fiber optic cable included in the HFD-S upgrade kit to create the MDS return link between the Systems cabinet (STIF board) and SSM.

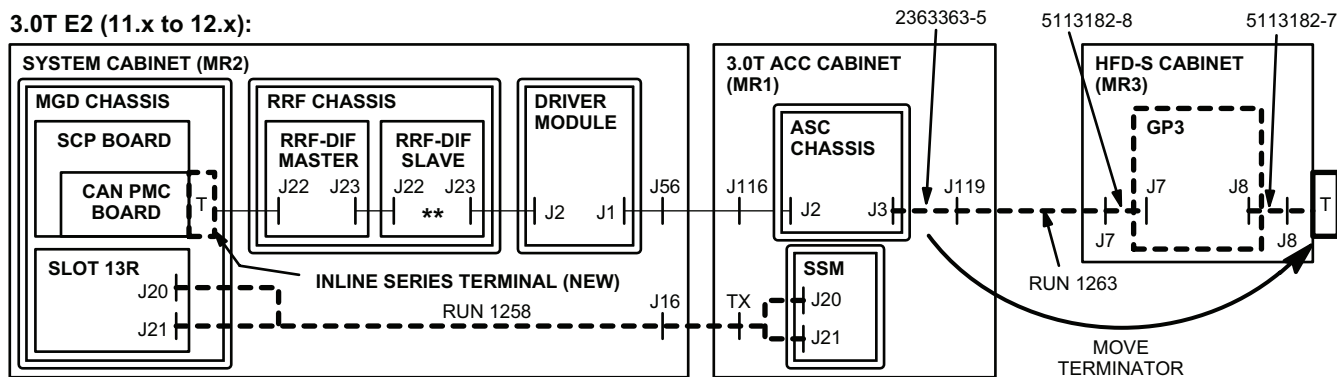
1.5T BRM/CRM/TRM (11.x to 12.x):



3.0T G3 (11.x to 12.x):



3.0T E2 (11.x to 12.x):



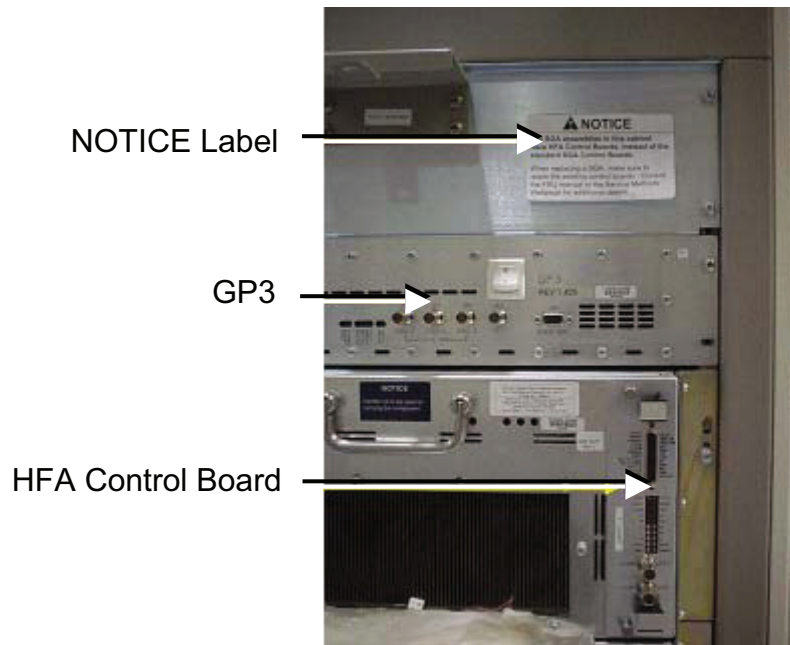
New Component: - - - - - Moved Component: ————

- * This cable, 2363363, was provided with the EXCITE I to EXCITE II upgrade.
- ** RRF-DIF SLAVE is only present systems with the 16 Channel option.

COMMUNICATION LINKS BEFORE AND AFTER HFD-S UPGRADE
ILLUSTRATION 3-7

3-7 Affix Labels

1. Attach the notice sticker to the front top panel of the HFD-S cabinet, in line with the HFA control boards. See Illustration 3-8.
2. Affix the new rating plate near to the ACGD rating plate that is currently in the cabinet.



ATTACH NOTICE LABEL TO FRONT TOP PANEL OF HFD-S
ILLUSTRATION 3-8

3-8 1.5T TRM Systems ONLY – Replace Gradient Filter

1. Replace the Gradient Filter per the procedure on the Service Methods Website under “Replacements” -> “Penetration Panel” -> “Gradient Filter Replacement Procedure”

3-9 Gradient Amplifier Power Supply Replacement

1.5T and 3.0T The procedure below needs to be followed if it has been determined, via the upgrade checklist, that the Gradient Power Supply needs to be replaced.

3-9-1 Personnel Requirements

Required Persons	Preliminary Reqs	Procedur e	Finalizatio n
---------------------	---------------------	---------------	------------------

1

3-9-2 Preliminary Requirements

3-9-2A Tools and Test Equipment

Item	Qty	Effectivity	Part#	Manufacture r
Universal Lift Hoist Kit with Gradient Attachments	1	-	46-317266G6, G7, G8, or G9	-
Phillips-head Screwdriver	1	-	-	-
Straight-edge Screwdriver	1	-	-	-
Digital Multimeter with high voltage probes	1	-	-	-

3-9-2B Safety

DANGER



FATAL ELECTRIC SHOCK HAZARD!!

THE GRADIENT CABINET/PDU CONTAINS HIGH VOLTAGE.

TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE YOU PERFORM THE FOLLOWING PROCEDURES. PERFORM LOCKOUT / TAGOUT PROCEDURE PER GE OSHA LOCKOUT / TAGOUT REQUIREMENTS LISTED IN THE FIELD SERVICE EHS MANUAL & TRAINING 2000 P/N 2135387-200.

WARNING



POSSIBLE PERSONAL INJURY.

HEAVY OBJECT - THE POWER SUPPLY WITH CASE WEIGHS 240 POUNDS/ 109 KG.

ASSISTANCE TO LIFT THIS PRODUCT (MECHANICAL OR ADDITIONAL PERSONNEL) IS REQUIRED.

3-9-3 Procedure

3-9-3A Amplifier Power Supply Replacement Procedure

1. Perform the [Gradient Lock Out Tag Out](#).

WARNING



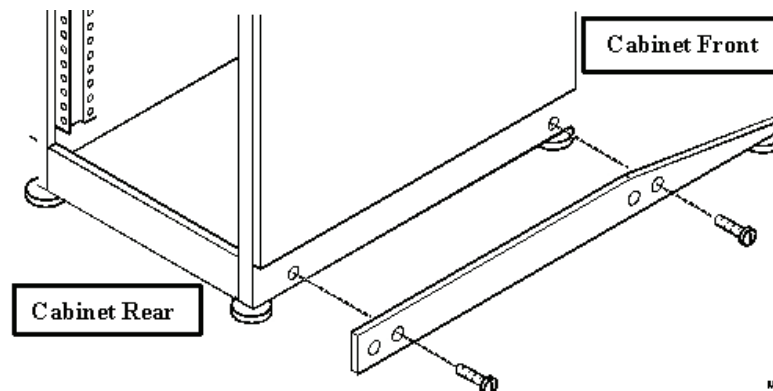
CABINET TIP HAZARD!

POSSIBLE PERSONAL INJURY!

CONFIRM THAT THE ALUMINUM ANTI-TIP LEGS STORED IN THE REAR OF THE CABINET ARE INSTALLED ON EITHER SIDE OF THE BASE OF THE CABINET BEFORE ATTEMPTING TO REMOVE A AMPLIFIER-POWER SUPPLY.

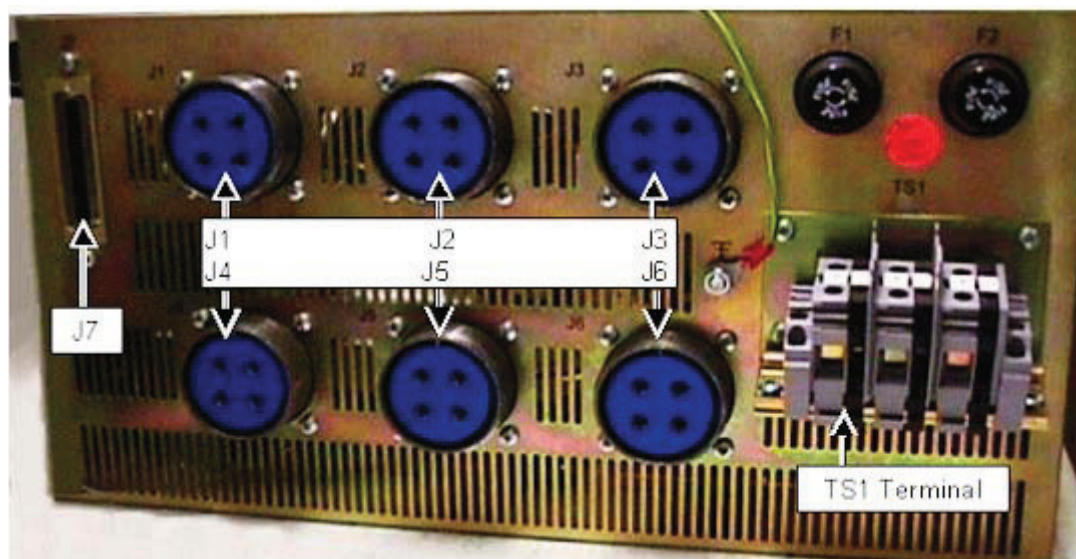
2. Using the screws included with the anti-tip legs, secure the aluminum anti-tip legs to the threaded holes on each side of the base of the Gradient Cabinet. See [Illustration 1](#).

Illustration 1: INSTALLING ANTI-TIP LEGS ON SIDES OF CABINET



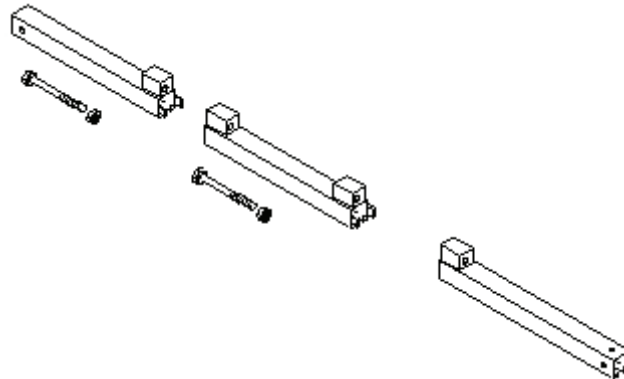
3. Remove the four pin connectors (J1, J2, J3, J4, J5 & J6) from the rear of the Amplifier Power Supply Module. See [Illustration 2](#).
3. Remove the data cable (J7) from the rear of the Amplifier Power Supply Module. See [Illustration 2](#).
4. Disconnect the TS1 terminal. See [Illustration 2](#).

Illustration 2: Amplifier POWER SUPPLY --REAR CONNECTIONS



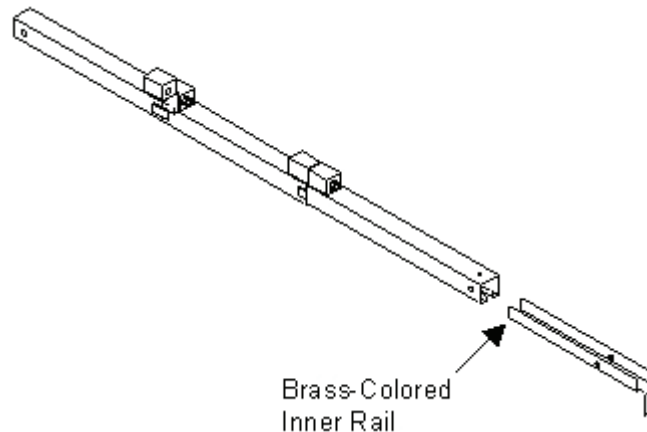
6. Locate the three (3) large Steel Rail sections in the Universal Hoist Kit and place them on a flat surface for assembly. See [Illustration 3](#).

Illustration 3: STEEL RAIL SECTIONS, HEX BOLTS, AND HEX NUTS



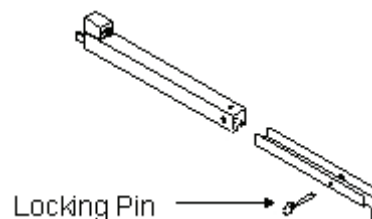
7. Arrange the rail sections in the proper order (rear, center, then front) and fasten them together with 4.75" X 1/2" caps crews and hex nuts.
8. Use the crescent wrenches to tighten the cap screws.
9. Locate the Brass Colored Inner Rail and insert it into the Steel Rails as shown in [Illustration 4](#).

Illustration 4: ASSEMBLED STEEL RAILS AND BRASS-COLORED INNER RAIL



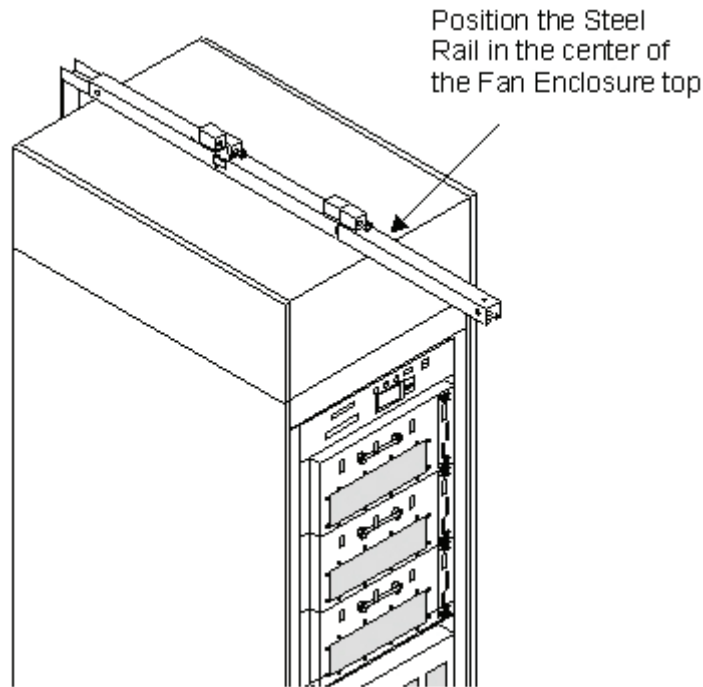
10. Insert the Brass Colored Inner Rail into the open end of the rear rail section.
11. Locate one of two (2) locking pins in the lift kit. Insert one locking pin into the hole near the open end of the rear rail section. This locking pin secures the Brass Colored Inner Rail to the outer Steel Rail. See [Illustration 5](#).

Illustration 5: REAR STEEL RAIL, LOCKING PIN, AND BRASS-COLORED INNER RAIL



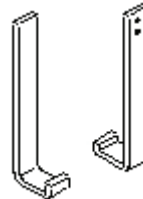
12. Place the Steel Rail assembly on top of the Fan Enclosure at the top of the Gradient Cabinet. Center the Rail Assembly on the top of the enclosure. See [Illustration 6](#).

Illustration 6: POSITIONING OF THE STEEL RAIL ASSEMBLY



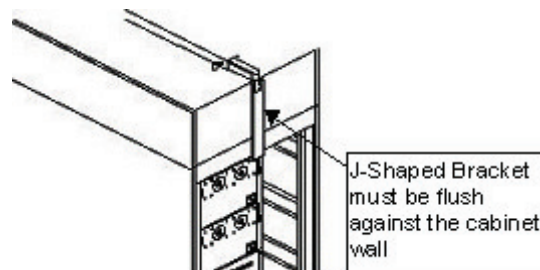
13. Locate the Brass Colored J-shaped Bracket in the Lift Kit. See [Illustration 7](#).

Illustration 7: J-SHAPED BRACKET



14. Open the rear door of the Gradient Cabinet. Observe how the J-shaped Bracket fits into the rear doorframe. See [Illustration 8](#).

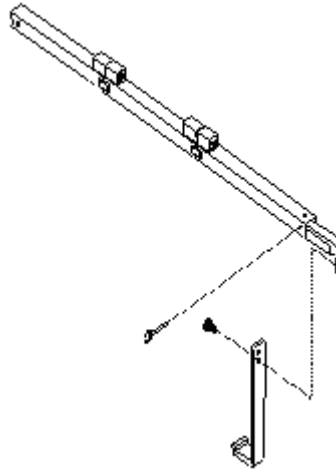
Illustration 8: POSITIONING OF THE J-SHAPED BRACKET



15. Slide the Rail Assembly toward the front of the Gradient Cabinet. Maintain its centered position.

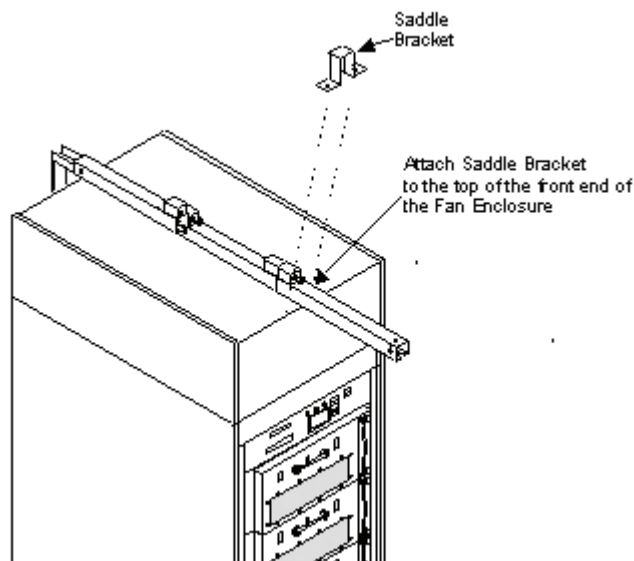
16. Attach the J-shaped Bracket to the Brass Colored Inner Rail so that the J-shaped Bracket is flush against the rear panel of the Fan Enclosure. Use the flat Phillips-head screws to secure the J-shaped Bracket to the Brass Colored Inner Rail. See [Illustration 9](#).

Illustration 9: CONNECTION OF THE J-SHAPED BRACKET TO THE STEEL RAIL ASSEMBLY



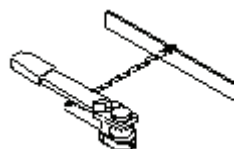
17. Once the J-Shaped Bracket is secured to the cabinet and the Steel Rail, attach the Saddle Bracket to the top of the Fan Enclosure. The screw holes are located at the front end of the Fan Enclosure. The Saddle Bracket prevents the Steel Rail from sliding from center position. See [Illustration 10](#).

Illustration 10: POSITIONING AND ATTACHMENT OF THE SADDLE BRACKET



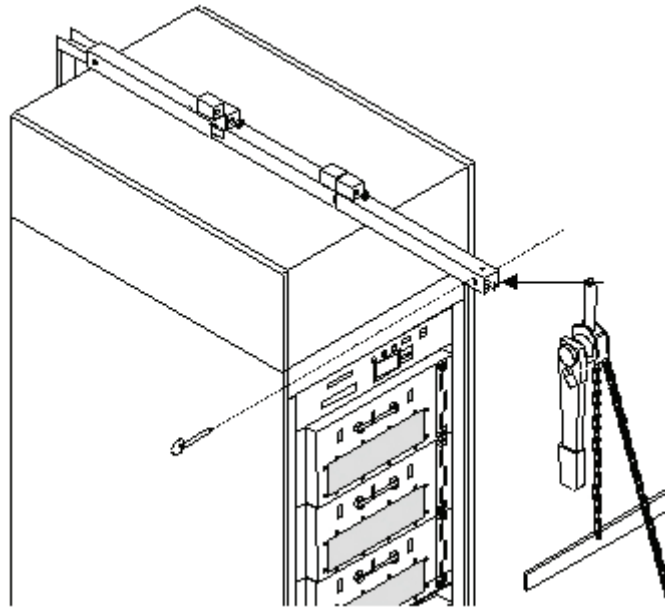
18. Locate the Hoist Assembly in the Lift Kit. See [Illustration 11](#).

Illustration 11: HOIST ASSEMBLY



19. Insert the Hoist Assembly into the open end of the front rail. Insert a locking pin into the holes near the front edge of the front rail. This locking pin prevents the Hoist Assembly from sliding out of the rails. See [Illustration 12](#).

Illustration 12: PLACEMENT OF HOIST ASSEMBLY INTO FRONT END OF STEEL RAIL ASSEMBLY



20. Remove the four (4) Phillips-head screws and washers from the Amplifier Power Supply (Amplifier-PS) front panel that secures the Amplifier-PS to the Gradient Cabinet. See [Illustration 13](#).

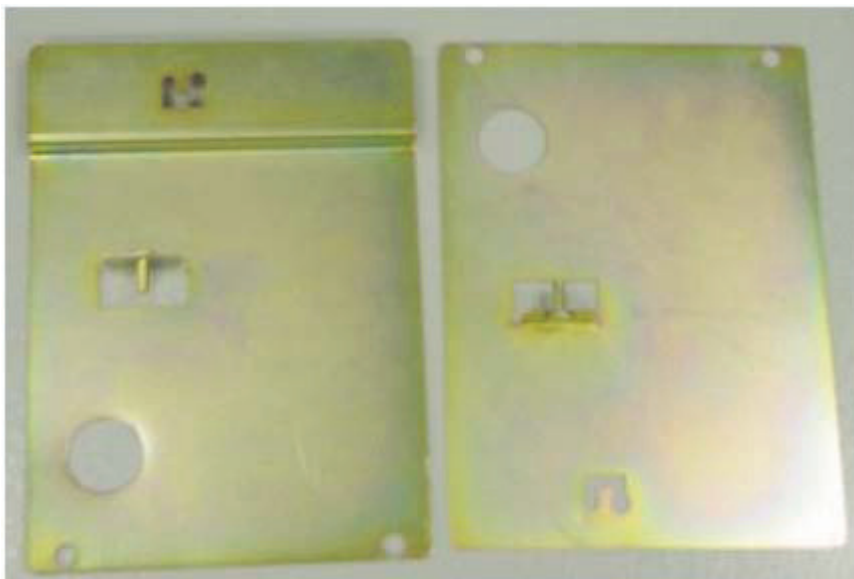
Illustration 13: SCREW LOCATION ON Amplifier-PS FRONT PANEL



21. Pull out the Amplifier-PS Module until its slide rails latch.

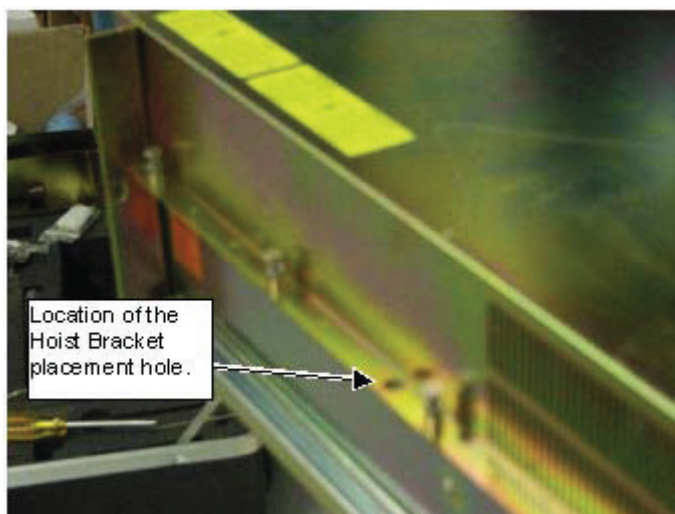
22. Remove the two Amplifier-PS brackets from the Universal Hoist Kit. See [Illustration 14](#).

Illustration 14: Gradient Amplifier-PS HOIST BRACKETS



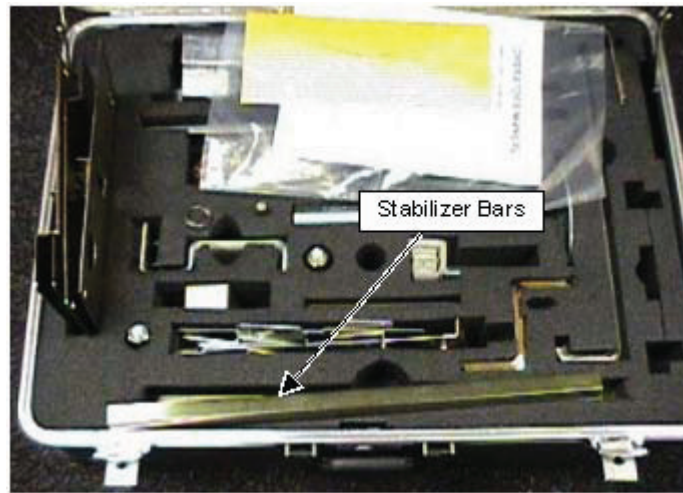
23. Brackets are labeled left and right. Place the appropriate bracket in the hole that is on the lip of sheet metal on each side of the Amplifier-PS. See [Illustration 15](#).

Illustration 15: LOCATION OF HOIST BRACKET PLACEMENT HOLE



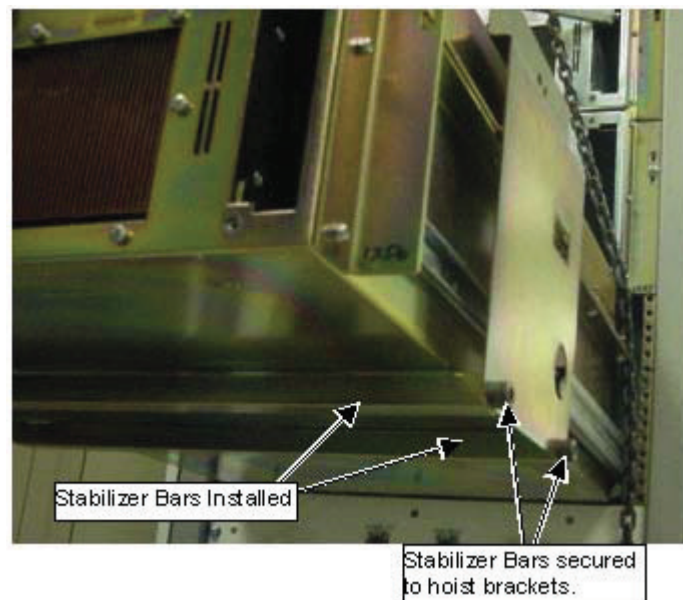
24. Locate the stabilizer bars in the Hoist Kit. See [Illustration 16](#).

Illustration 16: LOCATION OF STABILIZER BARS IN THE HOIST KIT



25. Using four screws from the hoist kit, attach the stabilizer brackets to the bottom of both hoist brackets. See [Illustration 17](#).

Illustration 17: ATTACHING STABILIZER BARS TO HOIST BRACKETS



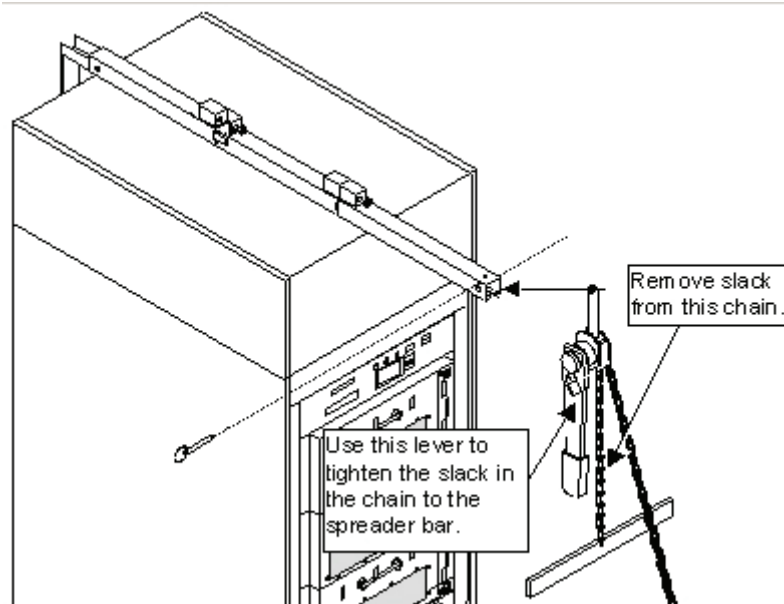
26. Attach the spreader bar of the hoist assembly to the Amplifier Hoist Brackets. This transfers the weight of the Amplifiers to the hoist. See [Illustration 18](#).

Illustration 18: ATTACHING THE SPREADER BAR TO THE HOIST BRACKETS



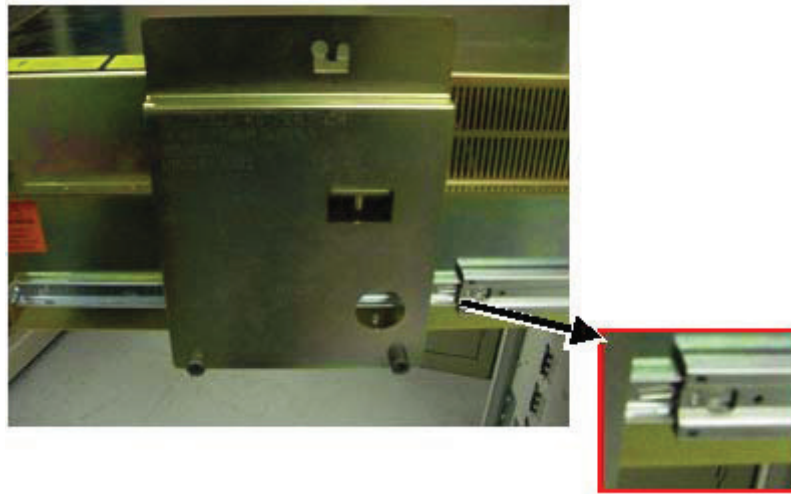
27. On the Hoist Assembly tighten the chain to the spreader bar to remove slack and support the weight of the Amplifier module. See [Illustration 19](#).

Illustration 19: TRANSFERING WEIGHT TO THE HOIST ASSMEBLY



28. Place the rail lever in the UP position on both sides of the Amplifier-PS. See [Illustration 20](#) for location of lever.

Illustration 20: RAIL LEVER LOCATION



29. Ensure that the entire weight of the Amplifier-PS is supported by the Universal Hoist.
30. Slide the module out and remove it from the rails that are attached to the cabinet.
31. Turn the module to the side and lower the module. See [Illustration 21](#).

Illustration 21: EXAMPLE OF A MODULE TURNED TO SIDE



32. Remove the spreader bar from the hoist brackets
33. Remove the spreader bar from the hoist brackets
33. Remove the hoist brackets from the failed module.

3-9-3B Module Replacement

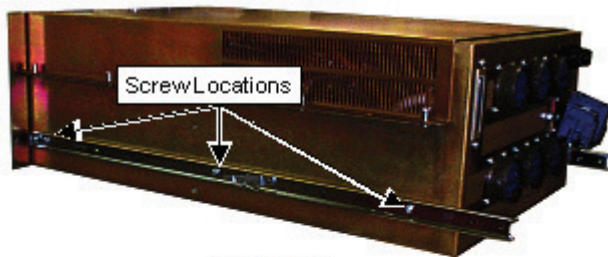
1. Remove shipping bands, etc. from replacement power supply crate and lift off the top.. See Illustration 22.



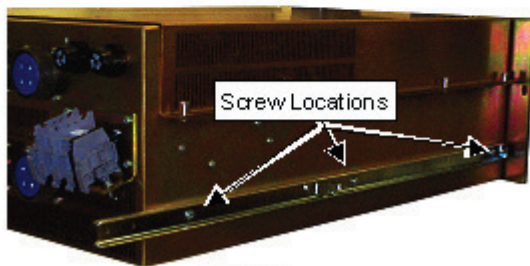
Illustration 22: ATTACHING RAILS TO Amplifier-PS

2. Remove the rails from the failed module and attach to the replacement module. See Illustration 23.

Illustration 23: ATTACHING RAILS TO Amplifier-PS



Right Side



Left Side



2. Attach the hoist brackets to the replacement module. See Illustration 24.

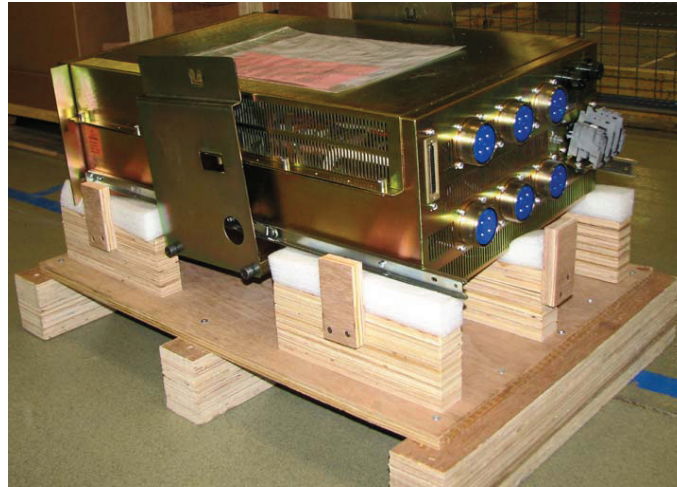
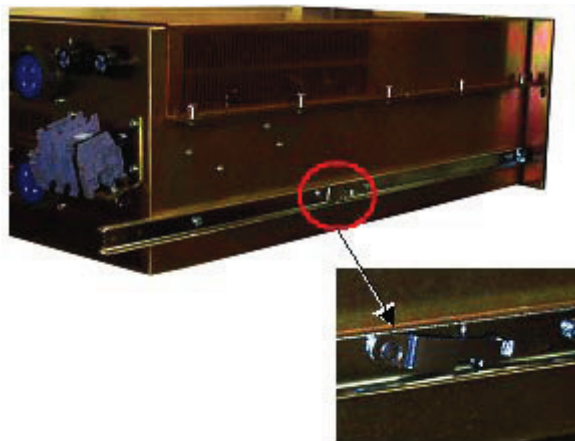


Illustration 24: ATTACHING HOIST BRACKETS TO POWER SUPPLY

3. Attach the spreader bar to the hoist brackets.
4. Fully extend the slide rail in the cabinet of the axis that is being replaced.
5. Raise the replacement Amplifier using the hoist assembly and move module in position to place rails into the slide mechanism on the cabinet.
6. Install the Amplifier-PS into the cabinet by inserting the module into the mating half of the rail assemblies.
7. Place the slide-rail lever in the DOWN Position. See Illustration 25.

Illustration 25: RAIL LEVER LOCATION



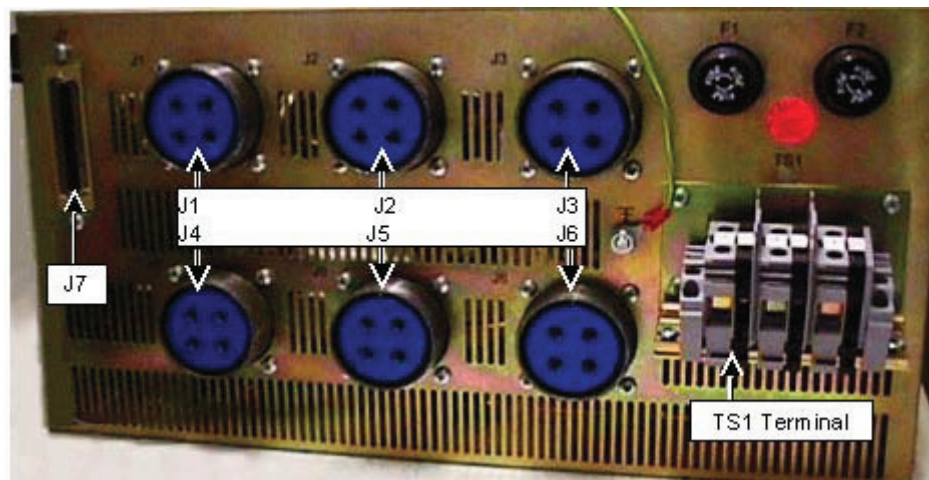
8. Push the Amplifier-PS assembly all the way into the cabinet.
9. Remove the spreader bar from the hoist brackets.
10. Remove the hoist brackets.
11. Secure the assembly to the front panel by replacing the four (4) Phillips-head screws and washers from the Amplifier-PS front panel. See Illustration 26.

Illustration 26: SCREW LOCATION ON Amplifier-PS FRONT PANEL



12. Replace the four pin connectors (J1, J2, J3, J4, J5 & J6) at the rear of the Amplifier-PS Module. See Illustration 26.
13. Replace the data cable (J7) at the rear of the Amplifier-PS Module. See Illustration 27.
14. Connect the TS1 terminal. See Illustration 27.

Illustration 27: Amplifier POWER SUPPLY --REAR CONNECTIONS



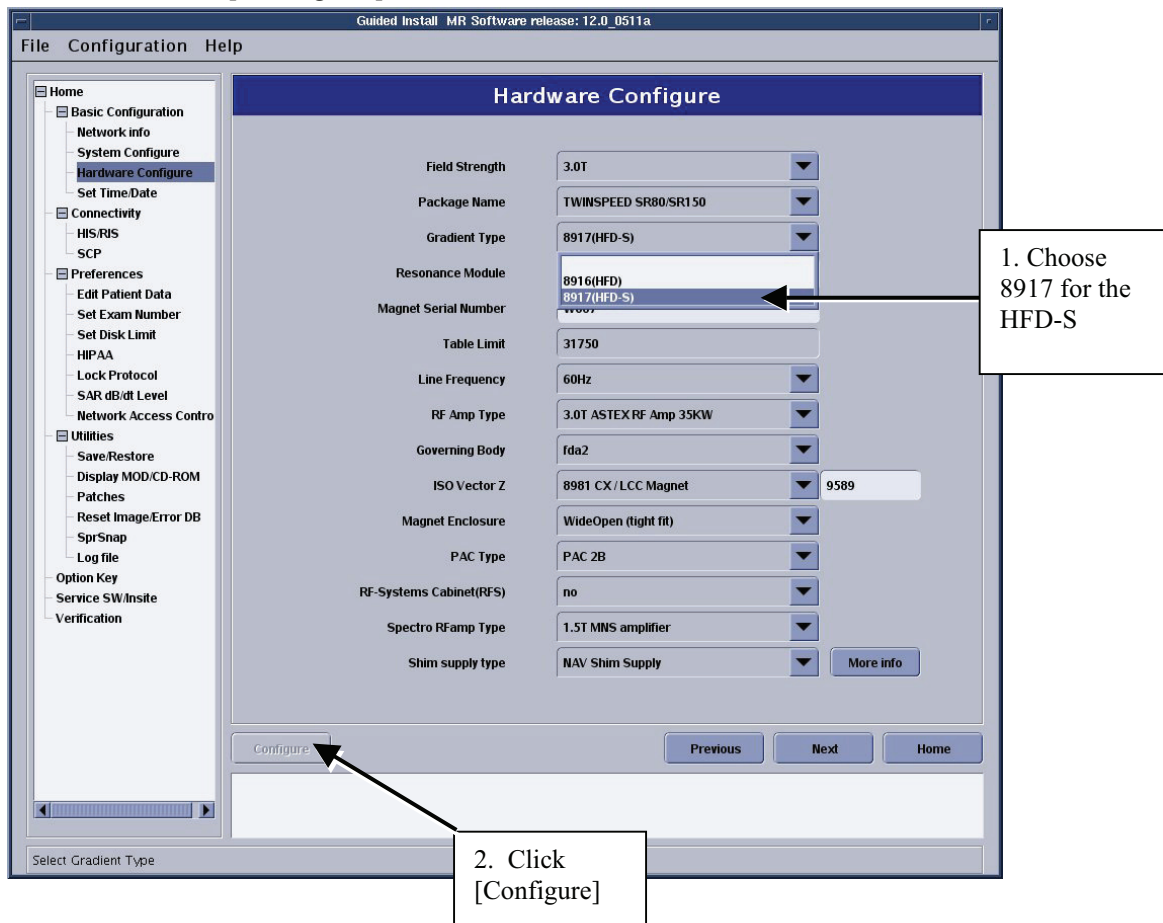
16. Remove the Universal Hoist from the cabinet and return to the kit.

3-10 LOTO Removal and Power-Up

1. Notify all personnel in the area that the system will be powered on.
2. Remove all LOTO devices and restore power to the MDP.
3. After reapplying power to the PDU, press the green reset button.
4. Power on the computer.
5. Power up the gradient cabinet
6. Set up for a scan to verify gradient driver comes up to Ready.

4- SOFTWARE INSTALLATION AND MODIFICATIONS

1. If upgrading from M3 (with ACGD) to M3A or M4 (with HFD-S), install the new software.
2. Open up the Guided Install (ID: **root**, PASSWORD: **operator**)
3. In guided install, click on the “Hardware Configure” tab. Change the gradient type to HFD-S by selecting 8917 from the pull down menu. After the gradient selection has been made, click [Configure]. See Illustration 4-1.

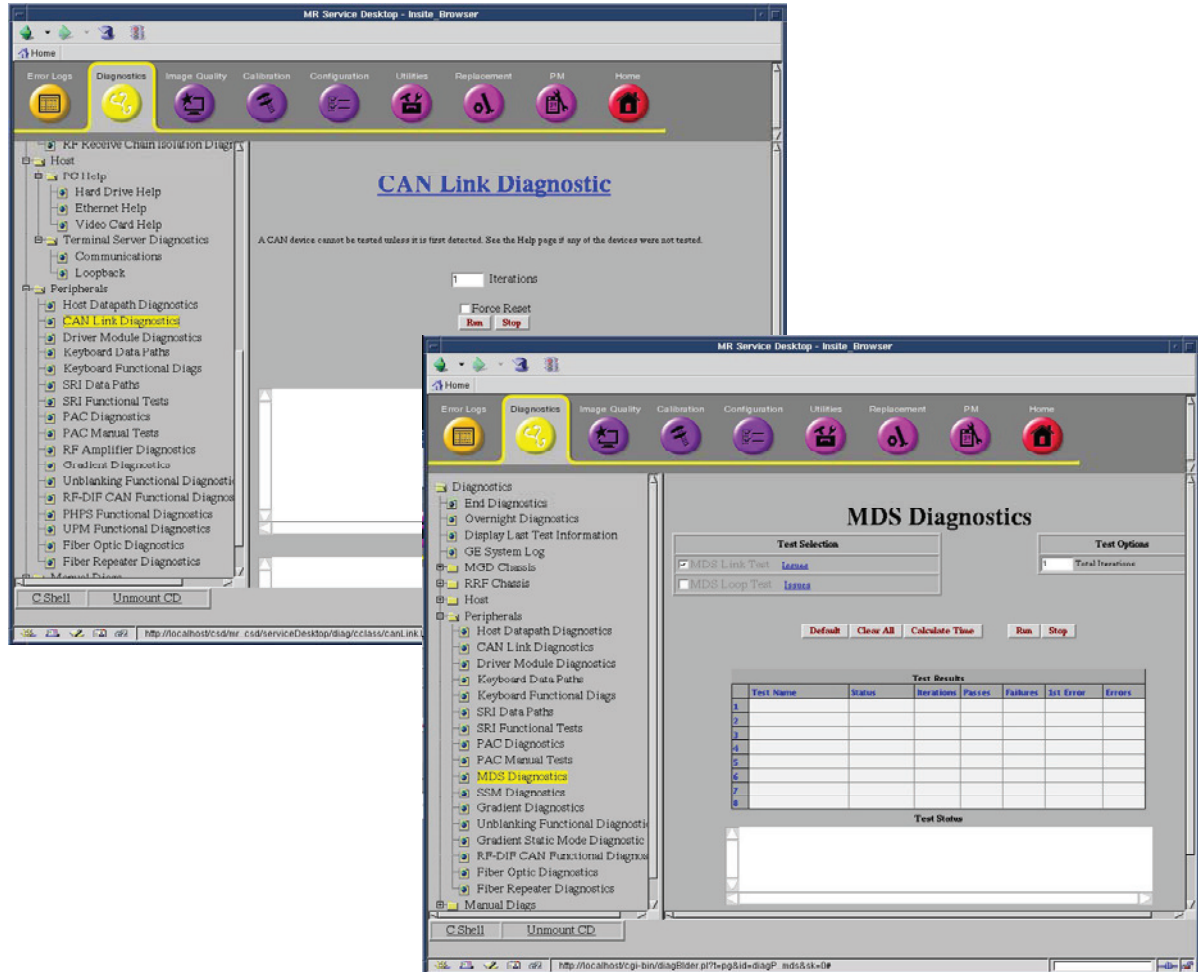


SELECTING 8917 FROM THE GUIDED INSTALL HARDWARE CONFIGURATION MENU
ILLUSTRATION 4-1

4. Reboot the system.

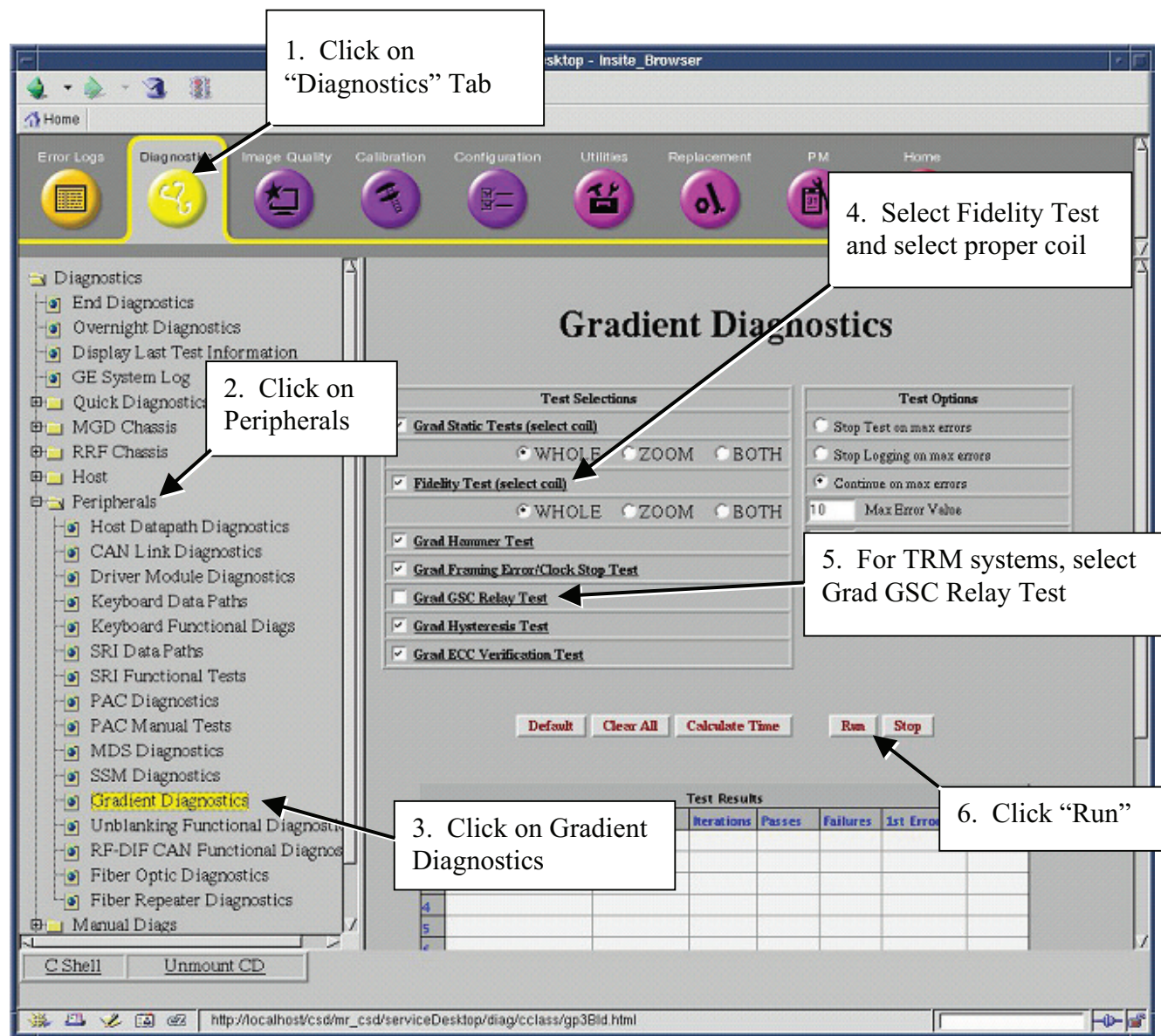
5- FUNCTIONAL CHECKS

1. From the Common Service Desktop, run the MDS link diagnostic and the CAN diagnostic tools to be sure that communication is functioning correctly. See Illustration 5-1.
2. If there are errors in either one of these tests, go through this procedure again to be sure that all of the MDS and CAN connections have been correctly routed and connected. . Also, please see the help information and diagrams available from the hyperlinks in the CAN Diagnostic page.



RUNNING MDS AND CAN DIAGNOSTICS
ILLUSTRATION 5-1

3. Run the Fidelity diagnostic. Also, for TRM systems, run the Grad GSC Relay test. See Illustration 5-2.



**RUNNING THE FIDELITY DIAGNOSTIC
ILLUSTRATION 5-2**

6- CALIBRATIONS

1. (1.5T TRM Only) Perform Geometry Checks
2. Run DC Offsets
3. Run Gradcal.
4. Run a full Grafidy 3 calibration.
5. Run full Planned Maintenance from PM Assist
 - a) PM LV Shim
 - b) PM SPT
 - c) PM EPI Cals
6. If site has the Probe option, perform Probe cals.

7- FINALIZATION

1. Perform SaveINFO with new HFD-S calibrations.
2. At the end of the upgrade of an ACGD cabinet to HFD-S for an EXCITE HD system, the following materials will be left unused:

Part Number	Description	M3000PT			M3000PW
		3.0T G3	3.0T E2	1.5T BRM/CRM	1.5T TRM
2363363-5	Cable, CAN, I/O J119 to ASC SLOT 18-J2	X		X	X
5113502	Run 1180 – MR3-J7 to MR2-J56	X	X		
5113502-8	Run 1263 - MR3-J7 to MR1-J119			X	X
5125863	Label, J119, E2 Accessory Cabinet IO Panel	X		X	X
5113182-6	Cable for grad switch internal GP3 to PDU		X		

3. Clean up all equipment and materials used during the installation
4. Return GP2. Use shipping label located in Appendix C of this manual.
5. Return the system to normal operating conditions for the customer.

APPENDIX A – INVENTORY LIST

M3000PT	
2331400	GP3 Assy for ACGD
2396050	HFA Control Board Assy
5132933	HFD-S Cable Kit
5113182-7	GP J8 to PDU J8
5113182-8	PDU J7 to GP J7
2363363-5	Cable, CAN IO J119 to ASC Slot 18-J2
5113502	Run 1180 MR3-J7 to MR2-J56
5113502-8	Run 1263 MR3-J7 to MR1-J119
5113182-6	PDU J5 to GP J9
5133379	CABLE (Run 1258 MR2 A11-J16 to MR1 TX to MGD Slot 13R-J20,J21 SSM J20, J21)
5125863	Label, 119, E2 Accessory Cabinet I/O
5125863	Label, 119, E2 Accessory Cabinet I/O
5125766	ACGD-S Label
5126918	PDU Backpanel ACGD, ACGD-S upgrade for Teal and Phoenix PDUs
2303317	CAN Termination Male
5117168	CAN Terminator Series
5134402	Fan FMI 60682 Kit
46-302200P3	Modified by Kit, Serial #, Model #, Description & Mfgd Date 2.25x1.70 Plate
M3000PW	
2331400	GP3 Assy for ACGD
2396050	HFA Control Board Assy
5132933	HFD-S Cable Kit
5113182-7	GP J8 to PDU J8
5113182-8	PDU J7 to GP J7
2363363-5	Cable, CAN IO J119 to ASC Slot 18-J2
5113502	Run 1180 MR3-J7 to MR2-J56
5113502-8	Run 1263 MR3-J7 to MR1-J119
5113182-6	PDU J5 to GP J9
5133379	CABLE (Run 1258 MR2 A11-J16 to MR1 TX to MGD Slot 13R-J20,J21 SSM J20, J21)
5125863	Label, 119, E2 Accessory Cabinet I/O
5125863	Label, 119, E2 Accessory Cabinet I/O
5125766	ACGD-S Label
5126918	PDU Backpanel ACGD, ACGD-S upgrade for Teal and Phoenix PDUs
2303317	CAN Termination Male
2382273	3T TRM Gradient Filter
5117168	Can Terminator, Series
5114402	Fan FMI 60682 Kit
46-302200P3	Modified by Kit, Serial #, Model #, Description & Mfgd Date 2.25x1.70 Plate

APPENDIX B – FAN MODIFICATION INSTRUCTIONS

B-1 EFFECTIVITY

TABLE 1
EFFECTIVITY SYSTEMS BY SERIAL NUMBER

00000210159MR0	00000201181MR5	00000204153MR1	00000206220MR6	00000207782MR4	00000209308MR6	00000209308MR6
00000220512MR8	00000201191MR4	00000204155MR6	00000206221MR4	00000207783MR2	00000209309MR4	00000209309MR4
00000199036MR5	00000202042MR8	00000204156MR4	00000206327MR9	00000207785MR7	00000209313MR6	00000209313MR6
00000206348MR5	00000202514MR6	00000204168MR9	00000206329MR5	00000207786MR5	00000209316MR9	00000209316MR9
00000206714MR8	00000202516MR1	00000204170MR5	00000206330MR3	00000207787MR3	00000209318MR5	00000209318MR5
00000194132MR7	00000202517MR9	00000204174MR7	00000206331MR1	00000207788MR1	00000209321MR9	00000209321MR9
00000194133MR5	00000202520MR3	00000204178MR8	00000206336MR0	00000207789MR9	00000209324MR3	00000209324MR3
00000194134MR3	00000202521MR1	00000204180MR4	00000206339MR4	00000207790MR7	00000209326MR8	00000209326MR8
00000198021MR8	00000202522MR9	00000204181MR2	00000206344MR4	00000207791MR5	00000209335MR9	00000209335MR9
00000198135MR6	00000202523MR7	00000204186MR1	00000206678MR5	00000207792MR3	00000209337MR5	00000209337MR5
00000198488MR9	00000202976MR7	00000204950MR0	00000206679MR3	00000207793MR1	00000209342MR5	00000209342MR5
00000198794MR0	00000202978MR3	00000204951MR8	00000206687MR6	00000207796MR4	00000209760MR8	00000209760MR8
00000198795MR7	00000202979MR1	00000204956MR7	00000206688MR4	00000208266MR7	00000209761MR6	00000209761MR6
00000198829MR4	00000202980MR9	00000205277MR7	00000206709MR8	00000208267MR5	00000209763MR2	00000209763MR2
00000198830MR2	00000202983MR3	00000205279MR3	00000206715MR5	00000208268MR3	00000210031MR1	00000210031MR1
00000199069MR6	00000203179MR7	00000205280MR1	00000206722MR1	00000208269MR1	00000210153MR3	00000210153MR3
00000199234MR6	00000203180MR5	00000205281MR9	00000207148MR8	00000208271MR7	00000210155MR8	00000210155MR8
00000199243MR7	00000203181MR3	00000205283MR5	00000207153MR8	00000208272MR5	00000210502MR1	00000210502MR1
00000199387MR2	00000203185MR4	00000205286MR8	00000207156MR1	00000208275MR8	00000210503MR9	00000210503MR9
00000199390MR6	00000203186MR2	00000205288MR4	00000207157MR9	00000208276MR6	00000210509MR6	00000210509MR6
00000199391MR4	00000203187MR0	00000205289MR2	00000207159MR5	00000208277MR4	00000210514MR6	00000210514MR6
00000199392MR2	00000203613MR5	00000205290MR0	00000207163MR7	00000208279MR0	00000210515MR3	00000210515MR3
00000199394MR8	00000203614MR3	00000205293MR4	00000207166MR0	00000208281MR6	00000210519MR5	00000210519MR5
00000199633MR9	00000203615MR0	00000205295MR9	00000207167MR8	00000208283MR2	00000210527MR8	00000210527MR8
00000199691MR7	00000203617MR6	00000205306MR4	00000207168MR6	00000208284MR0	00000210535MR1	00000210535MR1
00000199692MR5	00000203620MR0	00000205572MR1	00000207169MR4	00000208285MR7	00000211477MR5	00000211477MR5
00000199693MR3	00000203621MR8	00000205576MR2	00000207171MR0	00000208286MR5	00000211481MR7	00000211481MR7
00000199694MR1	00000203622MR6	00000205580MR4	00000207172MR8	00000208289MR9	00000211484MR1	00000211484MR1
00000199696MR6	00000203623MR4	00000205582MR0	00000207175MR1	00000208295MR6	00000211627MR5	00000211627MR5
00000199698MR2	00000203624MR2	00000205583MR8	00000207176MR9	00000208303MR8	00000211628MR3	00000211628MR3
00000199700MR6	00000203625MR9	00000205586MR1	00000207769MR1	00000208306MR1	00000211629MR1	00000211629MR1
00000200334MR1	00000203626MR7	00000205587MR9	00000207771MR7	00000208309MR5	00000211630MR9	00000211630MR9
00000200340MR8	00000203627MR5	00000205589MR5	00000207772MR5	00000208311MR1	00000211631MR7	00000211631MR7
00000200342MR4	00000203628MR3	00000205592MR9	00000207774MR1	00000208312MR9	00000211632MR5	00000211632MR5
00000200343MR2	00000203631MR7	00000205596MR0	00000207776MR6	00000208317MR8	00000211633MR3	00000211633MR3
00000200345MR7	00000203633MR3	00000205599MR4	00000207777MR4	00000209100MR7	00000211634MR1	00000211634MR1
00000200864MR7	00000203635MR8	00000206206MR5	00000207778MR2	00000209287MR2	00000211649MR9	00000211649MR9
00000201173MR2	00000203637MR4	00000206207MR3	00000207779MR0	00000209304MR5	00000211650MR7	00000211650MR7
00000201177MR3	00000204147MR3	00000206211MR5	00000207780MR8	00000209306MR0	00000211651MR5	00000211651MR5
00000201178MR1	00000204152MR3	00000206214MR9	00000207781MR6	00000209307MR8	00000211654MR9	00000211654MR9

00000211658MR0	00000213333MR8	00000215019MR1	00000216590MR0	00000218354MR9	00000221081MR3	00000222332MR9	00000224637MR9
00000212033MR5	00000213339MR5	00000215021MR7	00000216593MR4	00000218356MR4	00000221082MR1	00000222335MR2	00000224638MR7
00000212034MR3	00000213340MR3	00000215051MR4	00000216594MR2	00000218362MR2	00000221083MR9	00000222545MR6	00000224639MR5
00000212035MR0	00000213343MR7	00000215146MR2	00000216595MR9	00000218414MR1	00000221084MR7	00000222546MR4	00000224641MR1
00000212036MR8	00000213346MR0	00000215152MR0	00000216954MR8	00000218417MR4	00000221085MR4	00000222547MR2	00000224642MR9
00000212037MR6	00000213347MR8	00000215157MR9	00000216961MR3	00000218418MR2	00000221086MR2	00000222548MR0	00000224644MR5
00000212038MR4	00000213348MR6	00000215158MR7	00000216962MR1	00000218659MR1	00000221321MR3	00000222550MR6	00000224646MR0
00000212040MR0	00000213376MR7	00000215164MR5	00000216963MR9	00000218662MR5	00000221323MR9	00000222551MR4	00000224651MR0
00000212043MR4	00000213377MR5	00000215167MR8	00000217070MR2	00000218663MR3	00000221325MR4	00000222552MR2	00000225035MR5
00000212044MR2	00000213378MR3	00000215465MR6	00000217071MR0	00000218665MR8	00000221326MR2	00000222553MR0	00000225037MR1
00000212046MR7	00000213379MR1	00000215725MR3	00000217072MR8	00000218668MR2	00000221327MR0	00000222556MR3	00000225040MR5
00000212278MR6	00000213380MR9	00000215729MR5	00000217073MR6	00000218800MR1	00000221328MR8	00000222557MR1	00000225042MR1
00000212284MR4	00000213381MR7	00000215770MR9	00000217075MR1	00000218801MR9	00000221706MR5	00000222564MR7	00000225043MR9
00000212288MR5	00000213383MR3	00000215776MR6	00000217077MR7	00000218803MR5	00000221707MR3	00000222565MR4	00000225045MR4
00000212289MR3	00000213387MR4	00000215777MR4	00000217078MR5	00000218804MR3	00000221708MR1	00000222880MR7	00000225050MR4
00000212292MR7	00000213388MR2	00000215846MR7	00000217205MR4	00000218941MR3	00000221709MR9	00000222920MR1	00000225052MR0
00000212293MR5	00000213391MR6	00000215849MR1	00000217206MR2	00000218949MR6	00000221711MR5	00000222924MR3	00000225627MR9
00000212298MR4	00000213393MR2	00000215852MR5	00000217208MR8	00000218950MR4	00000221712MR3	00000222995MR3	00000225691MR5
00000212299MR2	00000213692MR7	00000215854MR1	00000217209MR6	00000219018MR9	00000221713MR1	00000222997MR9	00000225692MR3
00000212301MR6	00000213693MR5	00000215957MR2	00000217211MR2	00000219019MR7	00000222037MR4	00000222998MR7	00000225693MR1
00000212302MR4	00000213695MR0	00000215958MR0	00000217219MR5	00000219024MR7	00000222038MR2	00000222999MR5	00000225697MR2
00000212311MR5	00000213696MR8	00000215966MR3	00000217220MR3	00000219225MR0	00000222039MR0	00000223003MR5	00000225699MR8
00000212314MR9	00000213697MR6	00000216103MR2	00000217222MR9	00000219229MR2	00000222041MR6	00000223004MR3	00000225704MR6
00000212318MR0	00000213700MR8	00000216104MR0	00000217365MR6	00000219231MR8	00000222042MR4	00000223211MR4	00000225706MR1
00000212320MR6	00000213703MR2	00000216105MR7	00000217368MR0	00000219235MR9	00000222043MR2	00000223213MR0	00000226641MR9
00000212322MR2	00000213706MR5	00000216106MR5	00000217370MR6	00000219237MR5	00000222046MR5	00000223214MR8	00000226642MR7
00000212323MR0	00000213709MR9	00000216107MR3	00000217371MR4	00000219242MR5	00000222206MR5	00000223286MR6	00000226697MR1
00000212965MR8	00000214268MR5	00000216108MR1	00000217373MR0	00000219765MR5	00000222209MR9	00000223457MR3	00000226698MR9
00000212969MR0	00000214270MR1	00000216109MR9	00000217467MR0	00000220508MR6	00000222221MR4	00000223468MR0	00000226699MR7
00000212973MR2	00000214271MR9	00000216455MR6	00000218078MR4	00000220509MR4	00000222222MR2	00000223472MR2	00000226700MR3
00000212975MR7	00000214273MR5	00000216459MR8	00000218079MR2	00000220510MR2	00000222273MR5	00000223473MR0	00000226704MR5
00000212976MR5	00000214567MR0	00000216460MR6	00000218106MR3	00000220511MR0	00000222274MR3	00000223476MR3	
00000212977MR3	00000214568MR8	00000216461MR4	00000218205MR3	00000220516MR9	00000222275MR0	00000223478MR9	
00000212979MR9	00000214569MR6	00000216462MR2	00000218206MR1	00000220990MR6	00000222280MR0	00000223479MR7	
00000212980MR7	00000214570MR4	00000216464MR8	00000218207MR9	00000220991MR4	00000222281MR8	00000224075MR2	
00000212981MR5	00000214675MR1	00000216565MR2	00000218208MR7	00000220992MR2	00000222282MR6	00000224452MR3	
00000212983MR1	00000214677MR7	00000216576MR9	00000218210MR3	00000220993MR0	00000222328MR7	00000224453MR1	
00000212984MR9	00000214679MR3	00000216577MR7	00000218275MR6	00000221078MR9	00000222329MR5	00000224454MR9	
00000213331MR2	00000215017MR5	00000216580MR1	00000218276MR4	00000221079MR7	00000222330MR3	00000224456MR4	
00000213332MR0	00000215018MR3	00000216583MR5	00000218353MR1	00000221080MR5	00000222331MR1	00000224634MR6	

B-2 FIELD SUPPLIED MATERIALS

Item	Qty
Step Stool	1
Standard Phillips Head Screwdriver	1
Small Phillips Head Screwdriver	1
11/32" and 9mm Socket Wrench	1
9/32" and 7mm Socket Wrench	1
Pair of Gloves	1
Digital Multimeter	1

B-3 PURPOSE

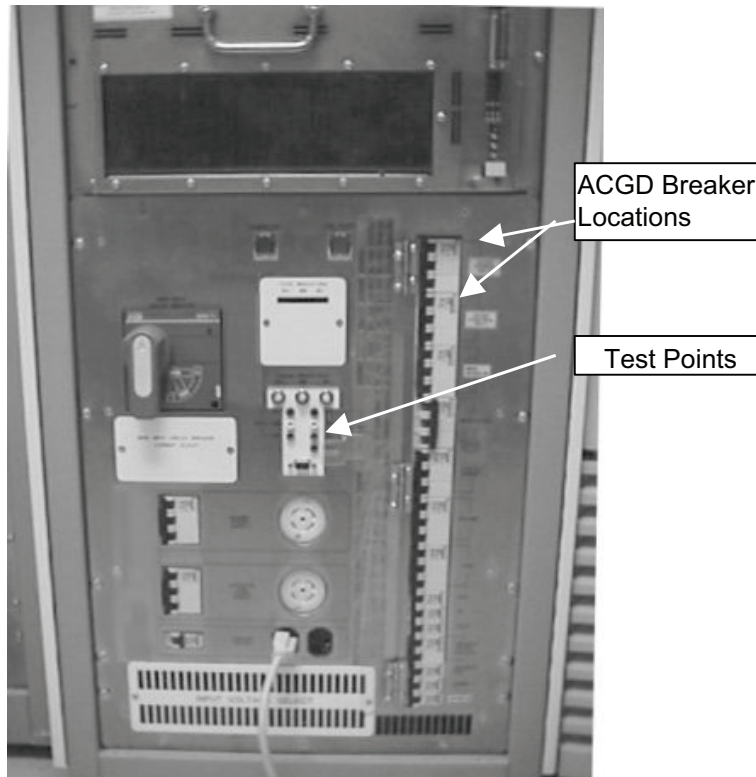
The purpose of Appendix B (FMI 60682) is to address the transistor-overheating problem at 50Hz sites. The existing rubber mounts from the fan assembly will be removed and replaced with nuts, washers and bolts to bring the fan closer to the ring increasing airflow.

B-4 PROCEDURE



FATAL ELECTRIC SHOCK HAZARD!! TO PREVENT FATAL ELECTRIC SHOCK, DISCONNECT POWER FROM THE PDU BEFORE YOU PERFORM THE FOLLOWING PROCEDURES. PERFORM LOCKOUT / TAGOUT PROCEDURE PER GE OSHA LOCKOUT / TAGOUT REQUIREMENTS LISTED IN THE *FIELD SERVICE EHS MANUAL & TRAINING 2000* P/N 2135387-200.

POWER DOWN THE ACGD CABINET BY TURNING OFF THE ACGD 208V CONTROL POWER AND THE ACGD 420 V POWER ON THE FRONT PANEL OF THE PDU. SEE ILLUSTRATION B-1.



**PDU (FRONT COVER REMOVED)
ILLUSTRATION B-1**

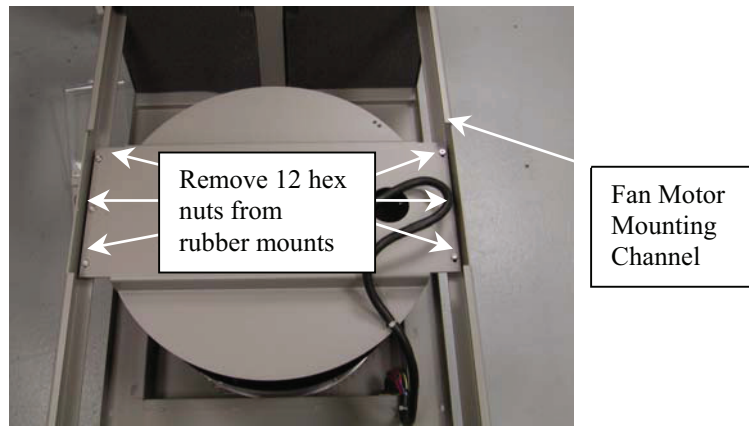
B-2.1 Ensure Proper LOTO

1. Ensure that system is locked out at the MDP per HFD-S upgrade requirements.
2. After power to the cabinet has been removed, set a digital multimeter to its highest AC Voltage range.

3. Verify that all energy has been dissipated. Measure power at the 420 and 208 test points on the front of the PDU.

B-2.2 Replacement Procedure Steps

1. Locate the Fan Enclosure at the top of the Gradient Cabinet. If working in the mobile environment, see van manufacturer specific procedure to remove fan motor/blower.
2. Remove two (2) Phillips-head screws from each fan assembly side panel.
3. Remove the side covers to provide access to the top grill for removal.
4. Remove the Fan Enclosure Top Grill.
5. Remove the six (6) 11/32" or 9mm hex nuts that secure the Fan Motor Mounting Channel to the Fan Enclosure. See Illustration B-2.



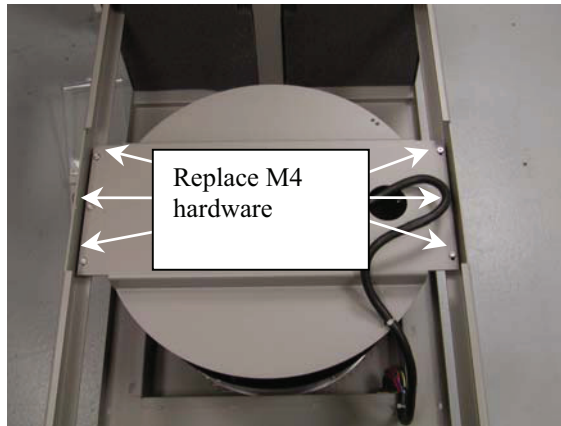
CABINET FAN MOTOR HOUSING AND FAN ASSEMBLY VIEWED FROM TOP
ILLUSTRATION B-2

6. Once hex nuts from rubber mounts have been removed, tilt fan up slightly and remove all six rubber mounts as shown in Illustration B-3.



TILTING FAN ASSEMBLY TO REMOVE RUBBER MOUNTS
ILLUSTRATION B-3

7. Replace the hardware with M4 screws, washers, and locknuts but **DO NOT TIGHTEN**. See Illustration B-4.

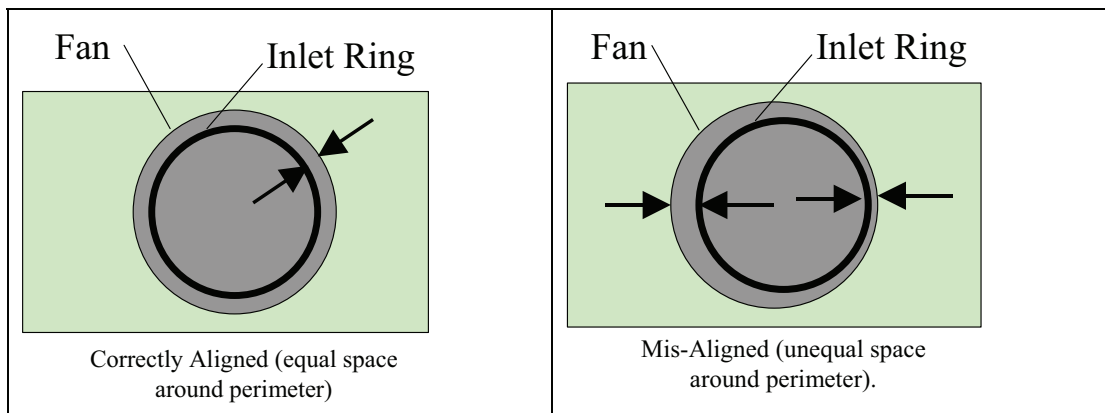


**GRADIENT CABINET FAN MOTOR HOUSING AND FAN ASSEMBLY VIEW FROM TOP
ILLUSTRATION B-4**

8. Center fan relative to inlet ring. Check for concentricity with fingers. See Illustration B-5.

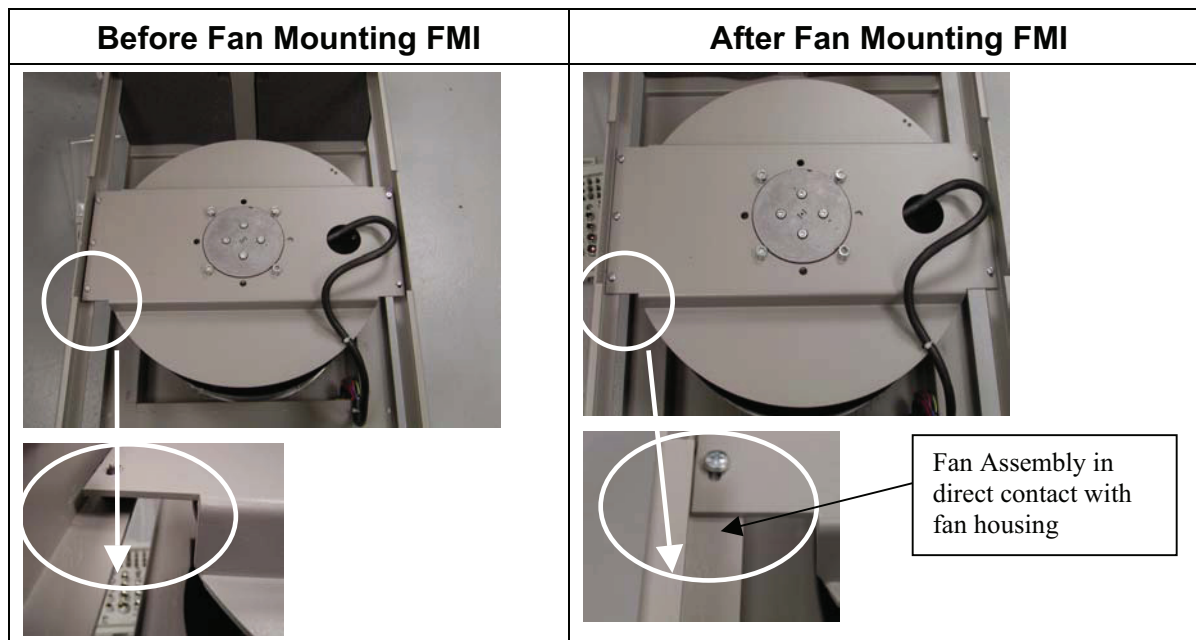


Personal injury may occur. The fan motor/blower assembly has sharp edges, wear protective gloves to avoid injury when removing from the fan assembly box.



**CHECKING FOR CORRECT SPACING AROUND FAN PERIMETER
ILLUSTRATION B-5**

9. Once fan has been correctly centered, tighten M4 hardware.
10. Once the hardware has been installed, the fan assembly should be in direct contact with the fan housing. See Illustration B-6.



BEFORE AND AFTER FAN HOUSING ASSEMBLY AS VIEWED FROM TOP

ILLUSTRATION B-6

11. Reverse the order of the above steps to reinstall the top and side covers.
12. Completing the steps in Appendix B qualifies for closure of FMI 60682. For countries equipped with Field MUST, GE CARES or similar on-line systems, use your laptop computer to file your service report.

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APPENDIX C – GOLDSEAL RETURN SHEET

Attention: GEMS GoldSeal

FDO: _____

Site Name: _____

Removal Date: _____

Field Engineer: _____